CERTIFICATE MATERIAL TEST REPORT

(DIN 50049.1 / EN 10204 3.1 / ISO 10474 3.1)

ISO 9001:2015

CAS METAL CO., Ltd.
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CERTIFICATE No.	: CAS250605-02	ORDER NO.	: PO755-005068
CUSTOMER	: NORCA INDUSTRIAL LLC	PROJECT NAME	: P.O. N8150
MATERIAL	: ASTM/ASME A/SA182F51/60(2023ED)	ISSUED DATE	: JUN. 05. 2025.

Heat No.	Spec	CHEMICAL COMPOSITION (%)																Non-destructive examination ③		
		C	SI	Mn	P	S	Ni	Cr	Mo	V	Cu	Nb/Cb	Al	N				UT ⑤	N/A	-
	Min.						4.5	22.0	3.0					0.14				MT ⑤	N/A	-
	Max.	0.030	1.00	2.00	0.030	0.020	6.5	23.0	3.5					0.20				PT ⑤	N/A	-
E131792	H.A. ①	0.017	0.45	1.30	0.022	0.001	5.56	22.21	3.04					0.185				Dimension & Visual Inspection	GOOD	
	P.A. ①																			

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※ Remark

* Country of Origin : KOREA * Country of Melt & Pour : KOREA * NACE MR0175 AND MR0103

* ISO 9001 : 2015 * PED 2014 / 68 / EU. * CORROSION TESTED TO ASTM A262 PRACTICE E : No Defect

ARTICLES	Q'ty	MECHANICAL PROPERTIES	TENSILE TEST ④ (Ø12.5 x 50L)						Hardness Test	CHARPY IMPACT TEST(J)				BEND TEST
			Location ②	Temp.	Y.S.	T.S.	E.L.	R.A.	HBW	Temp. : 40 °F				Radius :
					N/㎟ (MPA)		%			Min. :				
		Lot No.		℃	MIN. 450	MIN. 655	MIN. 25	MIN. 45	MAX 220	Avg. : 1	2	3	AVG	Angle :
09) B16.5 150# SO RF 8"	10EA													
10) B16.5 150# SO RF 12"	40EA													
11) B16.5 150# SO RF 14"	5EA	CAS25-N24	T		491	686	42	63	162, 160	91	89	102	94	
12) B16.5 150# SO RF 16"	15EA													
13) B16.5 150# SO RF 18"	5EA													
14) B16.5 150# SO RF 20"	5EA	Simulation P.W.H.T.	#1											
15) B16.5 150# SO RF 24"	5EA		#2											
16) B16.5 150# SW RF S/40S 1"	20EA		#3											

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Heat Treament ④											
	Normalizing	°C	HRS		Tempering	℃	HRS		Quenching	℃	HRS
	Solution	1040 °C	6.0 HRS	W.C	Stablization	℃	HRS		Annealing	℃	HRS

☐ **Witnessed By**

☐ **Reviewed By**

We hereby certify that the material herein has been made and tested accordance with the above specification and also with the requirement called for by the above order.

CAS METAL Co., Ltd. QUALITY CONTROL DEPART

Manager of Quality Assurance Dep't

Notes
① - H.A. = Heat Analysis, P.A. = Product Analysis
② - location : L = Longitudinal Tr = Transverse T = Tangential R = Radial Treated S.R. = Stress Relieved
③ - Surface Condition : A.F. = As Forged R.M. = Rough Machined F.M. = Finish Machined
④ - A.C. = Air Cooled, F.C. = Furnace Cooled, W.C. = Water Cooled, O.C. = Oil Cooled
⑤ - UT = Ultrasonic Test, MT = Magnetic Test, PT = Penetrant Test

ACCEPTED

BY JR DATE 1/19/26

Job : 089-25 PO: 40301 Line: 002 HT: E131792